

SALES PERFORMANCE ANALYSIS USING MICROSOFT EXCEL

Interactive Dashboard and Business Intelligence Report

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1. Executive Summary

Overview

This project presents an end-to-end sales analytics solution developed entirely in Microsoft Excel. Three independent datasets—Orders, Customers, and Products—were integrated into a single master dataset using relational lookup functions.

The project applies data cleaning, feature engineering, PivotTables, PivotCharts, KPI reporting, and dashboard development to transform raw sales data into meaningful business insights. The resulting dashboard enables users to monitor revenue, profitability, customer behavior, product performance, and operational efficiency through an interactive interface.

The objective is to demonstrate practical business analytics and Excel dashboard development skills commonly used by business analysts, financial analysts, and actuarial professionals.

2. Introduction

Organizations generate thousands of sales transactions every day. However, raw transactional data provides little value unless it is cleaned, organized, and analyzed.

Microsoft Excel remains one of the most widely used analytical tools across finance, consulting, insurance, and business functions. This project demonstrates how Excel can be leveraged to perform data preparation, exploratory analysis, KPI reporting, and dashboard creation without the need for specialized BI software.

The dashboard developed in this project provides management with a clear overview of sales performance and supports informed business decision-making.

3. Problem Statement

Businesses often maintain transactional, customer, and product information in separate datasets. This separation makes it difficult to obtain a consolidated view of organizational performance.

The primary challenge addressed in this project is to integrate these datasets into a unified analytical model and develop an interactive dashboard capable of answering key business questions related to sales, profitability, customer behavior, and product performance.

4. Objectives

The primary objectives of this project are:

- Clean and validate raw sales datasets.
 - Merge customer, product, and order information into a master dataset.
 - Generate business metrics through calculated fields.
 - Analyze revenue and profitability across products, categories, occasions, and cities.
 - Develop an interactive Excel dashboard using PivotTables and PivotCharts.
 - Present actionable business insights for management.
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5. Dataset Description

The project utilizes three related datasets.

Orders Dataset

The Orders dataset contains transaction-level information including Order ID, Customer ID, Product ID, Quantity, Order Date, and Delivery Date.

Customers Dataset

The Customers dataset contains customer-specific information including Customer ID, Customer Name, Gender, and City.

Products Dataset

The Products dataset contains product-level information including Product ID, Product Name, Category, Occasion, and Selling Price.

These datasets are linked using Customer ID and Product ID to create a unified master dataset.

6. Data Cleaning & Preparation

Prior to analysis, several preprocessing steps were undertaken to ensure data quality.

Data Cleaning Steps

- Converted all datasets into structured Excel Tables.
- Checked for duplicate records.
- Validated missing values.
- Standardized text formatting.
- Verified date consistency.
- Confirmed data relationships across datasets.
- Merged customer and product information into the orders dataset using XLOOKUP.
- Created a consolidated master dataset for analysis.

These steps improved data consistency and ensured accurate reporting.

7. Feature Engineering

Several calculated variables were developed to facilitate business analysis.

Feature	Formula
Revenue	Quantity $\times$ Price
Estimated Cost	Price $\times$ Quantity $\times$ 70%
Profit	Revenue – Estimated Cost
Profit Margin	Profit $\div$ Revenue
Delivery Days	Delivery Date – Order Date
Month	TEXT(Order Date,"mmmm")
Quarter	Derived from Order Date
Year	YEAR(Order Date)
Weekday	TEXT(Order Date,"dddd")
Order Size	High / Medium / Low

These engineered variables enable profitability analysis, operational monitoring, and time-series reporting.

8. Dashboard Development

An interactive dashboard was developed using Microsoft Excel.

The dashboard includes:

- Total Revenue KPI
- Total Profit KPI
- Total Orders KPI
- Average Order Value KPI
- Profit Margin KPI
- Revenue by Occasion
- Revenue by Category
- Monthly Sales Trend
- Revenue by City
- Profit by Category
- Top Selling Products
- Order Size Distribution

Interactive slicers were incorporated to allow users to filter results by Occasion, Category, City, Month, and Gender.

9. Business Findings

Overall Business Performance

- The company generated **total revenue of ₹3,520,984** from **1,000 customer orders**.
 - Total estimated **profit amounted to ₹1,056,295**, resulting in an overall **profit margin of approximately 30%**.
 - The **average order value** was **₹3,521**, indicating a healthy average customer spend.
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Occasion-wise Performance

The analysis of sales by occasion revealed clear differences in customer purchasing behavior.

- **Anniversary** generated the highest revenue (**₹674,634**), making it the strongest-performing occasion.
- **Raksha Bandhan** ranked second with **₹631,585** in revenue.
- **All Occasions** and **Holi** also contributed significantly with revenues of **₹586,176** and **₹574,682**, respectively.
- **Diwali** generated the lowest revenue (**₹313,783**) among the major occasions analyzed.

Business Insight:

Marketing campaigns and promotional offers should be prioritized around **Anniversary** and **Raksha Bandhan**, where customer demand is strongest.

Category Performance

Product category analysis highlighted considerable differences in revenue generation.

- **Colors** was the highest revenue-generating category with **₹1,005,645** in sales.
- **Soft Toys (₹740,831)** and **Sweets (₹733,842)** were the next best-performing categories.
- **Plants (₹212,281)** and **Mugs (₹201,151)** generated comparatively lower revenues.

Business Insight:

Increasing inventory and promotional activities for the **Colors**, **Soft Toys**, and **Sweets** categories could improve overall sales performance.

Profitability Analysis

Since estimated costs were assumed to be **70% of selling price**, profit followed a similar pattern to revenue.

- **Colors** generated the highest estimated profit (**₹301,693**).
- **Soft Toys (₹222,249)** and **Sweets (₹220,153)** also delivered strong profitability.
- **Mugs** produced the lowest estimated profit (**₹60,345**).

Business Insight:

The business should continue investing in high-performing categories while evaluating strategies to improve the profitability of lower-performing categories.

Monthly Sales Trends

Sales varied considerably throughout the year.

- **August** recorded the highest revenue (**₹737,389**).
- **February** ranked second with **₹704,509**.
- **March** generated **₹511,823**.
- **January** reported the lowest monthly revenue (**₹95,468**).

Business Insight:

Sales demonstrate noticeable seasonal variation. Understanding the reasons behind strong performance in **August** and **February** can help improve marketing strategies during weaker months.

Weekly Sales Analysis

Revenue distribution by weekday showed interesting customer purchasing behavior.

- **Sunday** generated the highest revenue (**₹642,469**).
- **Saturday** ranked second (**₹521,219**).
- **Thursday** and **Friday** also recorded strong sales.
- **Tuesday** produced the lowest weekday revenue (**₹423,119**).

Business Insight:

Weekend shopping activity is considerably higher than weekday activity. Businesses may benefit from scheduling promotional campaigns or launching new products during weekends.

Delivery Performance

The overall average delivery time was approximately **5.53 days**.

Among occasions:

- **Holi** orders had the longest average delivery time (**5.78 days**).
- **Diwali** orders had the shortest average delivery time (**4.95 days**).

Among product categories:

- **Cake** products required the longest average delivery time (**6.02 days**).
- **Plants** had the shortest average delivery time (**5.11 days**).

Business Insight:

Delivery performance remained relatively consistent across categories and occasions, although perishable products such as cakes required slightly longer fulfillment times.

Order Value Distribution

Customer orders were classified into three categories.

Order Size Number of Orders

Medium	387
Low	356
High	257

Business Insight:

Most customer purchases fall within the **Medium** order-value segment, suggesting opportunities to encourage upselling and cross-selling to convert medium-value customers into high-value customers.

Overall Conclusion

The dashboard demonstrates that the business performs particularly well during **Anniversary** and **Raksha Bandhan** occasions, with the **Colors** category being the strongest contributor to both revenue and profit. Sales are highest during **August**, and customer purchasing activity peaks on **weekends**, especially **Sunday**. While overall delivery performance is consistent, targeted promotional strategies and inventory planning for high-performing categories and peak sales periods could further improve business performance.

10. Business Insights

The dashboard enables decision-makers to identify key business trends.

Examples include:

- Identification of the highest revenue-generating occasions.
- Comparison of profitability across product categories.
- Recognition of top-performing products.
- Analysis of customer demand across cities.
- Evaluation of monthly sales trends.
- Assessment of delivery performance.
- Identification of high-value customer orders.

These insights support strategic planning and operational improvements.

Note: Replace these example insights with findings from your completed dashboard.

10. Conclusion

This project demonstrates how Microsoft Excel can be used as a comprehensive business analytics tool.

Through data cleaning, integration, feature engineering, visualization, and dashboard development, raw transactional data was transformed into actionable business intelligence.

The project highlights the practical application of Excel for sales reporting and demonstrates analytical techniques that are widely used in finance, consulting, insurance, and business analytics.

11. Skills Demonstrated

Technical Skills

- Microsoft Excel
- PivotTables
- PivotCharts
- XLOOKUP
- Excel Tables
- Conditional Formatting
- Dashboard Design
- Data Cleaning

Analytical Skills

- Business Analytics
 - Sales Analysis
 - Financial Analysis
 - KPI Development
 - Profitability Analysis
 - Data Visualization
 - Decision Support
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References

- Microsoft Excel Documentation
- Microsoft Support – PivotTables & PivotCharts
- Microsoft Support – XLOOKUP Function
- Publicly available retail sales datasets (for educational purposes)